

gRPC MACH Architecture

TCO Networking Costs Calculator

Point-to-Point Integration Pattern · Protocol Buffer IDL · AWS Egress Pricing

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File 'complexUnaryRPCPattern.proto' uploaded successfully!



Phase 4 — TCO Networking Costs Report

Full cost breakdown including architecture tactics, unit economics, and stakeholder summary.

SERVICE UNDER ANALYSIS

order-management-system

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BUC1

1200 base RPS

200 End Users

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APPLIED ARCHITECTURE TACTICS

Client-side LB Server-side LB TLS OAuth 2.0 + JWT

Applied Architecture Tactics & Patterns

INFORMATIONAL — NO RPS IMPACT

TACTIC	VALUE	DESCRIPTION	RPS IMPACT
Client-side Load Balancing	—	Balances the emission of gRPC packets across server instances.	None
Server-side Load Balancing	—	Balances the reception of gRPC packets across backend replicas.	None
TLS (one-way)	0 reconnects/hr	TLS 1.3 encrypts every gRPC message. 30: Max Payload overhead: bytes per message — RFC 8446 Because of RFC 8446 Section-5.2 * TLSCiphertext	None

TACTIC	VALUE	DESCRIPTION	RPS IMPACT
		<i>outer header: 5 bytes: * + 1-byte opaque_type=23 (uint8 ContentType) * + 2-byte legacy_record_version (uint16 ProtocolVersion) * + 2-byte length (uint16 length). * TLSInnerPlaintext prefix (AAD): * + 1-byte Content type and no padding. * AEAD Pre-record nonce: RFC 8446 Section-5.3 * + 8-byte length (64 bits explicit portion added during encryption). * AEAD tag: RFC 5116 Section-5.1 * + 16-byte tag appended to encrypted_record for AEAD_AES_128_GCM ciphertext is exactly 16 octets longer than its corresponding plaintext . Handshake is connection-scoped — negligible RPS impact at this reconnect rate. AWS ACM certificates are free.</i>	
OAuth 2.0 + JWT — bearer token	Unary interceptor TTL 3600s 1 client(s)	<i>JWT Bearer header RFC 7515 Section-7.1 BASE64URL(UTF8(JWS Protected Header)) '.' * BASE64URL(JWS Payload) '.' * BASE64URL(JWS Signature) adds 650 bytes typical due to Payload overhead RFC 7519 §3 and RFC 9101: 650 bytes per message: Algorithm Standard claims RS256 Header 80 Payload (compact/std/ext) 200/400/600 Signature 344 Dots 2 Total 626/826/1026 → 650 typical . The token travels in every request's Authorization metadata header. Response messages carry no JWT overhead. Validation enforced by the configured gRPC interceptor.</i>	None



Security Byte Overhead Applied to Message Sizes

TLS OVERHEAD / MESSAGE	JWT OVERHEAD / REQUEST	EFFECTIVE REQUEST SIZE
30 bytes (typical)	650 bytes (typical)	8705 bytes
30 / bytes typical	300 / 650 / 1200 bytes (min/typical/max)	
EFFECTIVE RESPONSE SIZE		
824 bytes		

MESSAGE TYPE	PROTO SIZE (BYTES)	EFFECTIVE SIZE (BYTES)	REQUESTS / MONTH	THROUGHPUT (GB/MONTH)	TRANSFER COST (USD/MONTH)
Request	8025	8705	3B	25,216.5198 GB	—
Response	794	824	3B	2,386.9514 GB	\$ 214.83

Total Cost of Ownership (TCO) Breakdown

Cost Component	DR / BC	Monthly Cost	% of TCO
AWS Egress (Response Transfer)	Networking	\$214.83	29.5%
Cloud Infrastructure ALB(s) \$40.88 ElastiCache \$242.36 Database/Backup \$2.30 Security Services \$68.00 Containerized Cluster \$146.00 API Gateway \$14.60	Cloud + Container	\$514.14	70.5%
FinOps Optimisation (RI/SP saving)	Cost Reduction	-\$25.23 saved	saving
Total Cost of Ownership (TCO)		\$703.74/mo	100.0%

* % of TCO is calculated against the sum of ALL cost rows shown above (must total 100%). DR = Disaster Recovery | BC = Business Continuity.



Unit Economics

TOTAL MONTHLY TCO (NETWORKING + CLOUD INFRA)

\$703.74

\$214.83 egress + \$203.37 cloud infra

 Per-Microservice Cost Metrics

TCO / MICROSERVICE / MONTH \$703.74 This service · full TCO	COST / REQUEST \$0.00000023 3.110.400.000 req/mo	ANNUAL PROJECTION \$8444.88 12× monthly TCO
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 Per-End-User Cost Metrics (200 end users)

TCO / END USER / MONTH \$3.5187 \$703.74 ÷ 200 users	TCO / END USER / DAY \$0.117290 Monthly ÷ 30	BUSINESS USE CASE — Pattern cost attribution
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 Revenue, ARPU & ROI Analysis

Revenue model: The value entered in Phase 1 is the *expected monthly revenue per end user* for this service. Total monthly revenue = that value × number of end users. **ARPU** = revenue per user per month. **ROI** = (total revenue – TCO) ÷ TCO. **Break-even users** = minimum end users needed to cover full TCO at current ARPU. Net Margin per user = ARPU – TCO per user. Present to Finance and Leadership to frame architecture cost as a revenue-generating investment.

TOTAL MONTHLY REVENUE \$3400 \$17.00/user × 200 users	ARPU / MONTH \$17 Revenue per end user / month	NET MARGIN / USER / MO \$13.4813 ARPU – TCO/user
MONTHLY ROI 713.0% (Revenue – TCO) ÷ TCO	ANNUAL ROI 713.0% 12× monthly projection	BREAK-EVEN USERS 25 users to cover full TCO
REVENUE / \$1 INFRA \$8.13 Revenue efficiency	ANNUAL REVENUE \$40.800 12× monthly revenue	

1 — IP Address Planning (CIDR Sizing)

The more redundancy you build into AWS (Multi-AZ, replicas, NAT Gateways, ENIs), the more private IP addresses are consumed. Plan your VPC CIDR block generously (e.g. /16 = 65,536 IPs) and **leave enough address space within each subnet CIDR for auto-scaling headroom**. A /24 subnet provides only 251 usable IPs — exhausted quickly in a multi-tier, auto-scaled architecture.

2 — Multi-Tier Architecture in AWS Terms

Separating MACH components (frontend, API gateway, gRPC services, databases) into **different subnets** makes this a **multi-tier architecture** in AWS terms. Each tier needs dedicated subnets with its own Security Groups and NACLs.

3 — Intra-VPC Connectivity

Use **VPC Endpoints** (PrivateLink) to access AWS services without internet routing; **Transit Gateway** for cross-VPC connectivity; **VPC Flow Logs** for visibility. Each component in a separate subnet requires explicit route table entries.

* Outbound (response) data transfer cost uses AWS standard internet egress tiered pricing for US East (N. Virginia): first 10 TB at \$0.09/GB, next 40 TB at \$0.085/GB, next 100 TB at \$0.07/GB, beyond 150 TB at \$0.05/GB. Inbound (request) traffic is free. Effective sizes include TLS (RFC 8446) and JWT (RFC 7519) overhead where applicable. [AWS Pricing Calculator](#) ↗